STAND-ALONE DEVICES/COMPUTERS

A stand-alone device or computer is one that can perform all needed operations without assistance or guidance from an external source. Imagine your phone in a place where there is no connection to any network. The phone is then in a stand-alone mode because it can only operate apps that do not need to connect to the internet to function. Games are mostly the kind of apps that can be used without connection to any network. You can also view previously saved photos and videos. About fifty years ago, computers are mostly operated as stand-alone. Once programs needed for word processing, graphics and accounting are installed in the computer, the only connection is to a printer to print output of these programs.

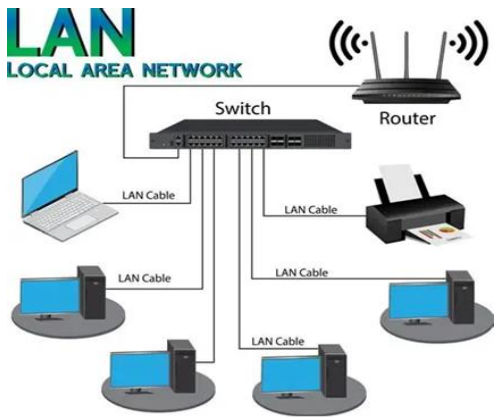
Obviously, no two computers can contain exactly the same thing especially user data. Sometimes an organization uses more than one computer, and it is desirable that what is in one computer is accessible to the other computers. There arose the need for computers to be connected to one another so that they can share resources. Computer A may have biodata information of employees while Computer B may have information on the salary structure and payment. Computer C may have a printer and/or scanner connected to it. The resources in this case are:

a. Biodata Information in Computer A
b. Salary Payment information in computer B
c. Printer / Scanner connected to Computer C.

From the simple example above, a resource can either be hardware (Printer/Scanner) or software (Information in A and B).

LOCAL AREA NETWORK (LAN)

Two or more computers connected together so as to share resources form a network. The computers or devices are usually within short range physical distances. However, a more advanced network involving many computers but still within short physical distances are referred to as Local Area Networks. Nowadays, connection of the computers to each other are through wires and wireless. Even though there are several ways the computers can be connected (topology), the most common type of topology is the Star topology. All devices/computers called nodes or clients, connect to a central device called Hub or Switch. In some cases the Switch could also be a server or a server could be one of the devices also connected to the Switch.

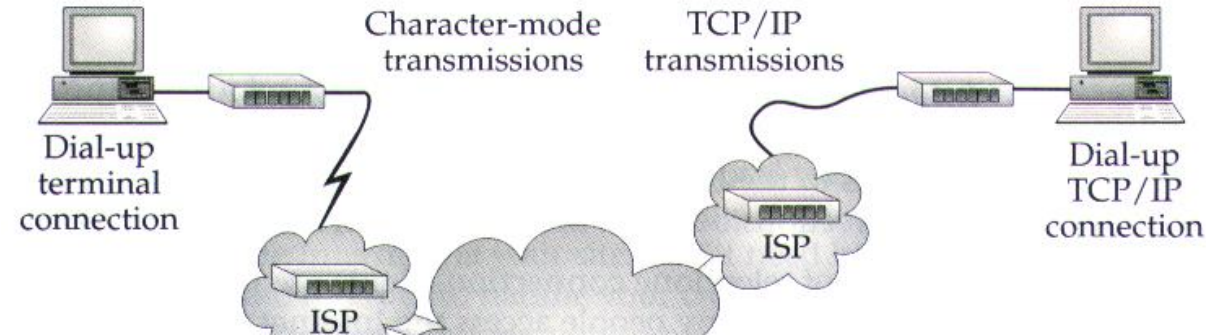


WIDE AREA NETWORK (WAN)

When nodes or clients of a LAN are situated at several kilometres or hundreds of kilometres, the normal connection methods will fail or provide unreliable service. Hence other connection methods are required. In many cases, there could be several LANs within a geographically wide area. These component LANs can be brought together under a single Wide Area Network. Examples of these are Campus Area Networks (CAN) and Metropolitan Area Networks (MAN). Ultimately, the overall connection of all networks all over the world is called INTERNET. The technologies with which each category of networks connect, varies from one to another. The major advantage of the INTERNET is that each node does not have to worry about how it connects to the other. In most cases, a node does not even know which other node it is connecting to. The "other node or nodes", are presented to a node as a service. For example, a person holding a smartphone (a node) connects to Google (a service) to ask for information. The Google server (another node) sends the answer to the smartphone without the smartphone knowing where the answer came from.

INTERNET

The Internet is known as "interconnection of computer networks". The Internet is a massive network of networks. It connects millions of computers together globally, forming a network in which any computer can communicate with any other computer as long as they are both connected to the Internet. It comprises of millions of computing devices that carry and transfer volumes of information from one device to the other. Desktop computers, mainframes, GPS



INTERNET / NETWORK PROTOCOL

Information that travels over the Internet does so via a variety of languages known as protocols. Basically, every device in a network is assigned an identity either temporarily during a connection or permanently, for several connections or for a very long time up to several years. The identity number usually called the Internet Protocol address (IP) consist of four parts separated by dots e.g. 192.168.0.1, 184.145.15.6. This number is translated by devices on the internet for several purposes. However, the most important thing to note is that the number must not have a duplicate in a particular network.

TCP/IP

The Transport Control Protocol (TCP) is the set of protocols by which data is transmitted across a network. In simple terms, TCP takes care of how data is transferred in a network.

- It breaks down the data into smaller packets that can be shared across a network effectively.
- At the receiver's end, TCP helps to arrange the data packets into a specific order to convey the initial information transferred through the web.
- To share the data packets, there will be a destination IP address.

At the early stage of Networking, the operators of computers will manually write commands using the IP addresses of other devices on the network in order to send or receive data or to make use of a resource. With much improvement, the identity of other clients on the network is masked by turning them into “services” . For example, if Mr A needs information, which is contained in Computer B, the information can be supplied to Mr A by a “service” computer which Mr A does not have to know the address. This protocol eliminates the need to always have to know or memorize IP addresses. This protocol is known as the World Wide Web.

Domain Name System (DNS)

With millions of computers and devices on the Internet, it becomes very difficult and cumbersome for a human being to remember IP addresses while operating in a network. The numbers are masked by giving Domain names to services on the internet. A protocol on the Internet Known as the DNS translates the domain name to the correct IP address internally so that the Internet identity is still maintained. For example, Mr Google can provide a service on the network and call it Google. To show that the service is commercial, it becomes Google.com

. to indicate that it will use a protocol called HyperText Transfer Protocol (HTTP) and also use World Wide Web, the full domain name becomes http://www.google.com

World Wide Web (WWW)

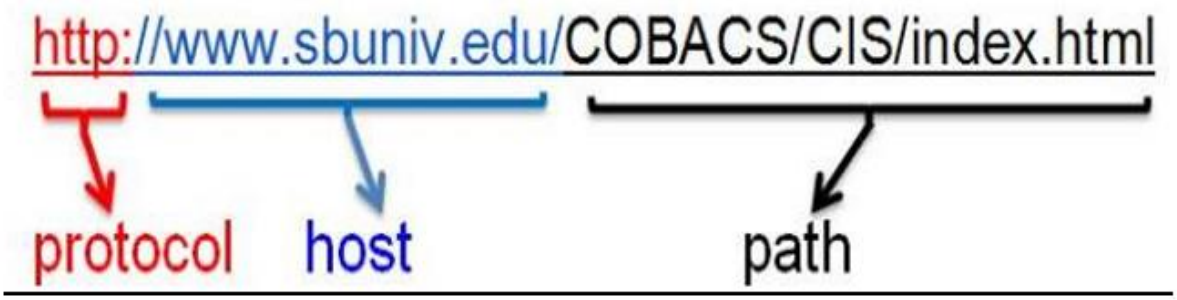
The World Wide Web (abbreviated WWW or the Web) is an information space where documents and other web resources are identified by Uniform Resource Locators (URLs), interlinked by hypertext links, and can be accessed via the Internet. The world wide web consists of billions of pages linked to each other that contain text, graphics, multimedia files, and other interactive software that are accessed using a browser. The term is often mistakenly used as a synonym for the Internet itself, but the Web is a service that operates over the Internet, just as e-mail also does.

The web works on three standards. These standards are generally adhere that make products that work with the World Wide Web.

URL (Uniform Resource Locator): These are the addresses that you enter into your web browser to connect to a web site. The URL is broken up into 4 parts which are the protocol, the hostname, the port number, and the path that you are requesting .

Protocol: The protocol part of an URL is the string of characters that you see before the hostname. Examples are http, ftp, telnet:, etc. They are separated from the hostname with a colon and two forward slashes (://). These protocols tell your browser what type of service to use when you connect with the web browser to the hostname. If you leave the protocol off your address, by default the Web Browser will assume you are using the HTTP protocol.

Hostname: The hostname is the address you are going to. Port Number: The port number is a number that you can append to the hostname with a colon (:) between them. This tell the port on the server computer. Path: This is the path on the server, culminating with the filename you are trying to reach. This path corresponds to an actual directory structure on the web server.



HTTP

Hyper Text Transfer Protocol (HTTP) The Hyper Text Transfer Protocol (HTTP) is the foundation of World Wide Web. Hypertext is well organized documentation system which uses hyperlinks to link the pages in the text documents. HTTP works on client server model. When a user wants to access any HTTP page on the internet, the client machine at user end initiates a TCP connection to server on port 80. When the server accepts the client request, the client is authorized to access web pages. To access the web pages, a client normally uses web browsers, who are responsible for initiating, maintaining, and closing TCP connections. HTTP is a stateless protocol, which means the Server maintains no information about earlier requests by clients.

WEB BROWSER

In order for the Web to work, web browsers and web servers which work hand in hand, are needed. The web browser is a piece of software, installed in client or user devices, that is used to interpret the information found in an HTML document, from a server. The browser displays he content of that document based upon the HTML tags found within it. When a web browser connects to a web server, the web server sends the requested document, if it exists, back to the web browser for display. In short, you type URLs into your browser or click a link, and the browser requests and displays those pages for you. Web browser can show text, audio, video, animation and more. It is the responsibility of a web browser to interpret text and commands contained in the web page. The various browser used mostly are Google chrome, Firefox, Opera, Edge, etc. Each web browser, besides providing the basic function of displaying the web page, also has various features like history, password maintaining, favorites, bookmarks etc.

WEB SERVER

A web server is a computer that stores HTML documents or scripts that generate html pages, otherwise known as web pages, and waits for connections from web browsers. The web server machine needs to be switched on, ready, and connected to the internet at all times. It is essentially waiting for an incoming request which could happen at any time. The web server program runs all the time, handling any incoming requests. For simple web pages, the web server program identifies a directory (aka a folder) as the web-root of the files to serve. The "path" part of the url maps into the web-root directory. The various server software used are Apache (free of charges) and IIS. Web server respond to the client request in either of the

following two ways: 1. Sending the file to the client associated with the requested URL. 2. Generating response by invoking a script and communicating with database.